Flight Test Specification & Test Plan

SQRT (Sub-zero Quantitative Research Team)

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Document History

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Author Contributions

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Key Reference Documents

1. UCD-DRD-2026-0

1 Executive Summary

The SQRT Flight Test Specification & Test Plan details the to planned full-system tests that the SQRT team will preform to verify the function of flight operations, including outlining test objectives, procedures, and outcomes. This plan will facilitate completion of testing required to ensure the TupperSat flies successfully and mission requirements are met.

2 Test Objective

The long-range test checks if the payload and ground station can talk to each other across a realistic distance. The six-hour bench test pushes the battery and the rest of the system to prove they'll hold up for the full mission time.

The six-hour bench test also includes verification of the experimental subsystem, ensuring proper valve actuation and successful camera recording during system operation. Of course, these tests don't copy every detail of flight. Things like altitude, temperature, or how the satellite moves in space aren't part of this round. We handle those environmental factors and other subsystem checks separately, using targeted tests and earlier validation work before we launch.

3 Test Procedure

1. Long-Range Communication Test

The purpose of this test is to verify the performance of the radio link over a representative operational distance.

- Fully assemble the TupperSat system and ground station.
- Ensure batteries are fully charged and the system is operating in normal mission mode.
- Deploy the ground station at a fixed location.
- Move the TupperSat transmitter progressively away from the ground station to increase separation distance.
- Continuously transmit telemetry packets at the nominal mission rate.
- Record received packets, RSSI values, and packet loss rate.
- At mid-test, trigger the experimental sequence to verify correct valve actuation and camera recording.
- Continue testing until the representative maximum distance is reached.

All data shall be logged for later analysis. Communication performance will be evaluated based on packet loss percentage and signal strength.

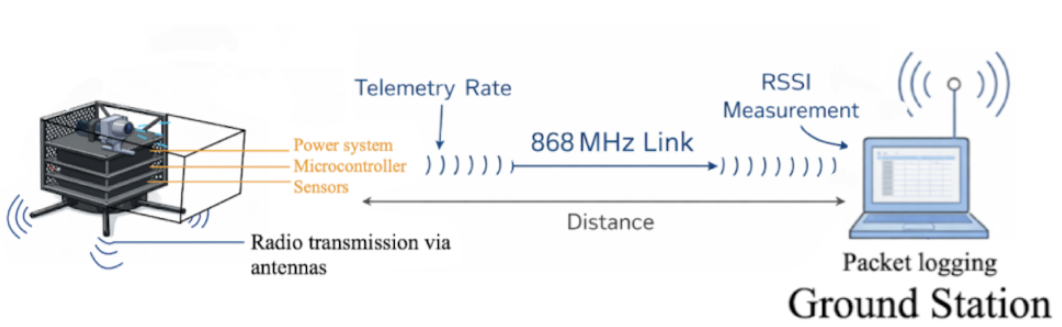


Figure 1: Long-Range Communication Test

2. Six-Hour Full System Bench Test

The purpose of this test is to simulate the full mission duration and verify system endurance.

- Fully charge the batteries before the test.
- Assemble the complete integrated system.
- Power on the system and place it in normal mission mode.
- Allow the system to run continuously for six hours.
- Monitor telemetry output, sensor data, and battery voltage throughout the test.
- Record any anomalies or interruptions.

The test concludes after six hours of continuous operation or when the system can no longer operate nominally.

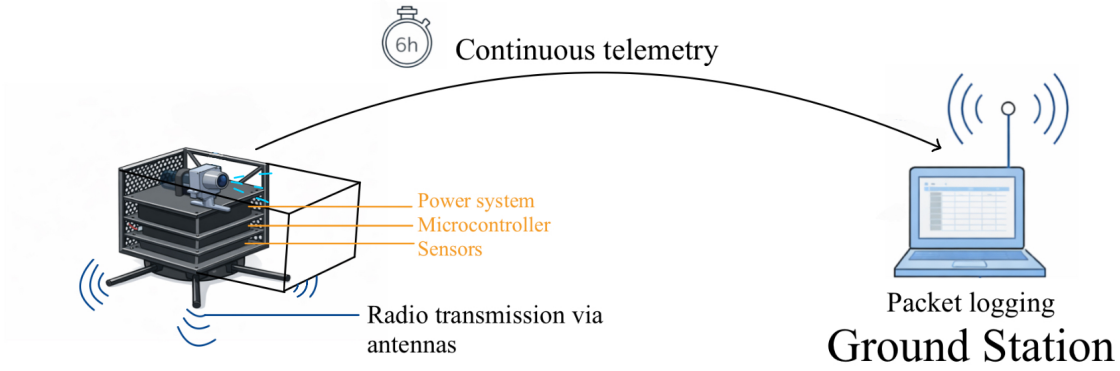


Figure 2: Six Hour System Bench Test

4 Expected Outcome

Table 1: Expected Outcome

Test	Objective	Pass Criteria	Fail Criteria
Long-Range Communication Test	Verify radio link performance over representative distance	Packet loss $< 20\%$ at max distance; RSSI above sensitivity threshold; continuous telemetry received	Packet loss $> 20\%$; sustained communication loss; corrupted telemetry
6-Hour Bench Test	Verify full system endurance and battery performance	Continuous operation for 6 hours; valid telemetry; successful valve actuation; successful camera recording; battery within safe limits.	Shutdown before 6 hours; battery below operational limit; telemetry interruption